Clustering agent representation for explaining reinforcement learning policies

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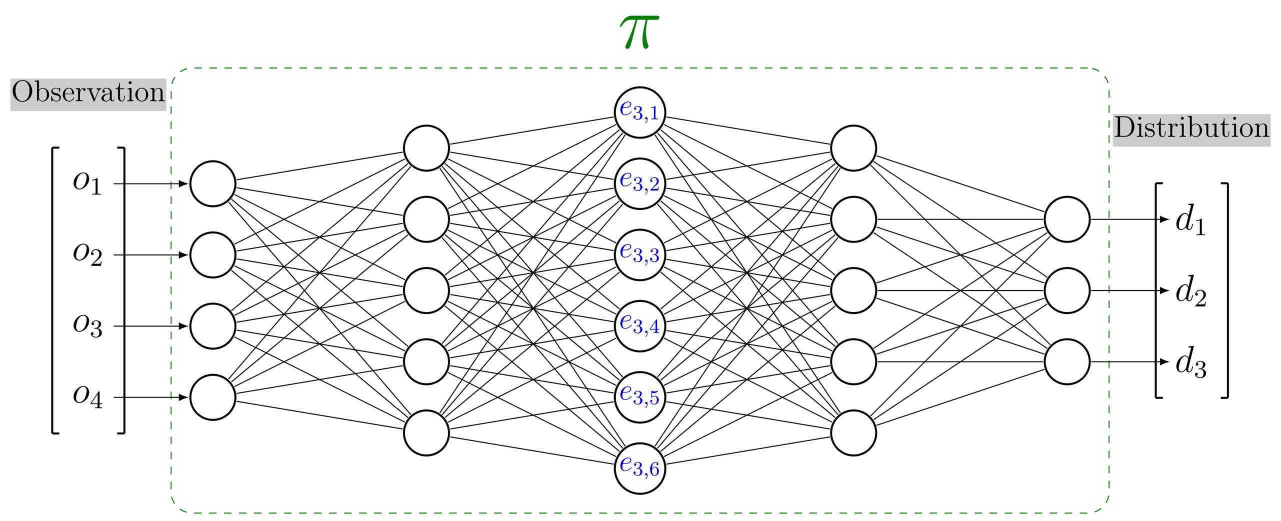
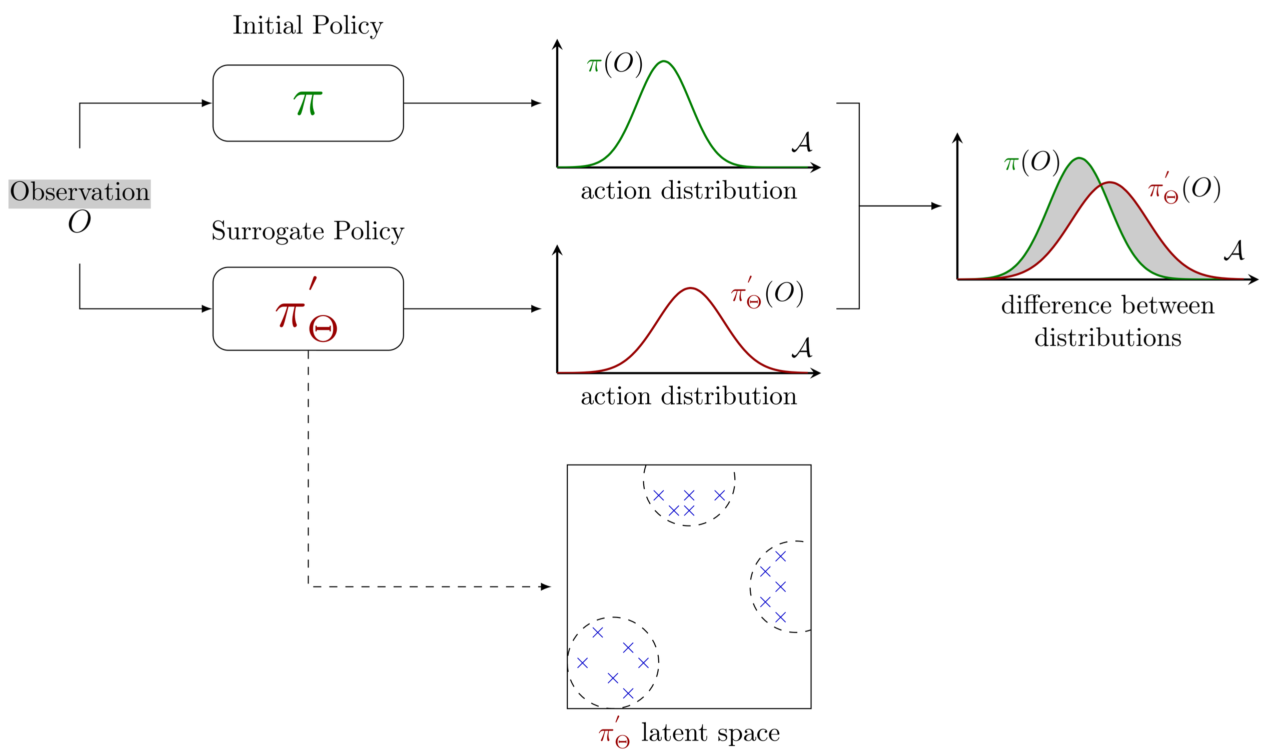
(1) Latent Space

Deep learning policies are represented as neural networks. Our goal is to study the neural network's latent space. This allows us to extract the semantics used for explainability.



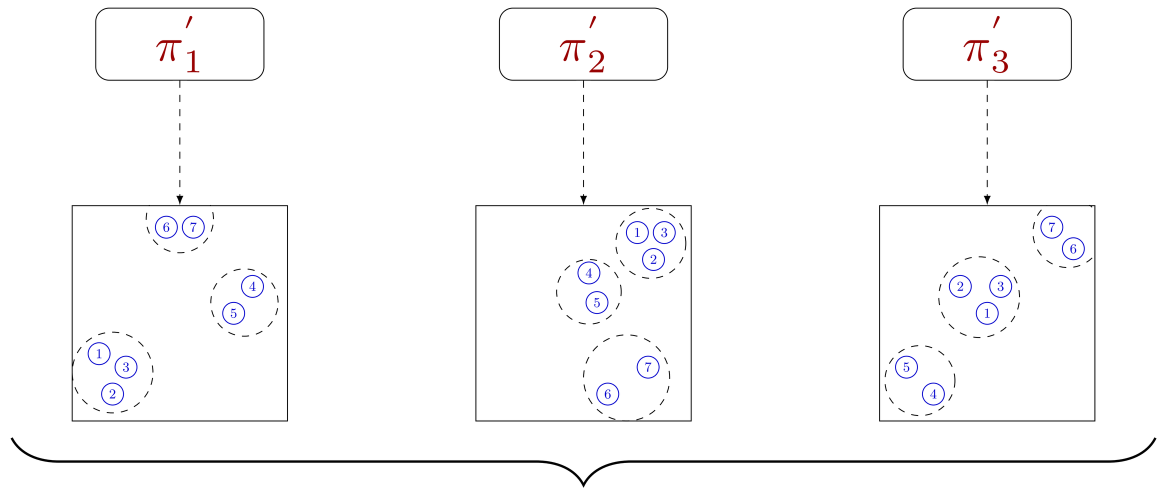
(3) Clustering Agent Representation

To ensure the representation is relevant for analysis, we repeat the latent space extraction process multiple times. This will allow us to confirm its stability before conducting comparisons between them.



(4) Clustering Universality

To compare whether latent spaces are similar, we use clustering analysis. Each latent space is processed with a clustering algorithm using N clusters. The Adjusted Rand Index is then used to measure the agreement between the clusterings, providing us with a stability metric.



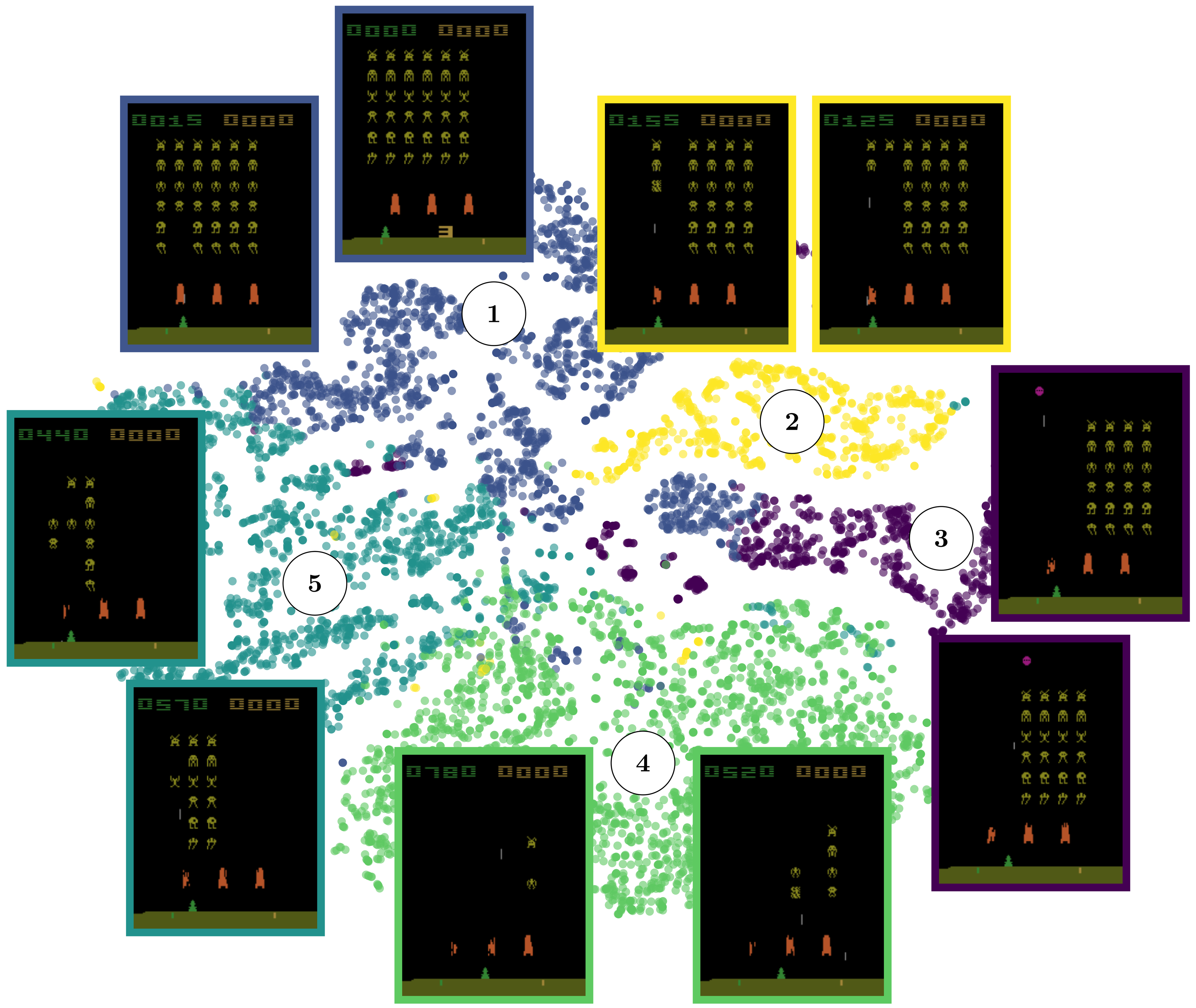
$$\textit{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^{n} \sum_{j=0}^{i-1} \mathrm{ARI}(\mathcal{C}_i, \mathcal{C}_j)$$

Abstract

Reinforcement learning has produced highly effective agents, but their decision-making processes often remain opaque, limiting their use in safety-critical applications. To address this, we propose a method using surrogate neural network policies to approximate agent behavior and analyze their latent space for clustering similar states. We introduce novel metrics to evaluate the stability and interpretive relevance of these clusters. Experiments on Atari environments demonstrate that this approach improves explainability. Our latent-space analysis reveals the underlying principles of complex agent behavior, advancing mechanistic interpretability in reinforcement learning.

(2) Semantic Latent Space

We observe that the distances between embeddings in the latent space are meaningful. The closer the embeddings are in the latent space, the closer they are semantically.

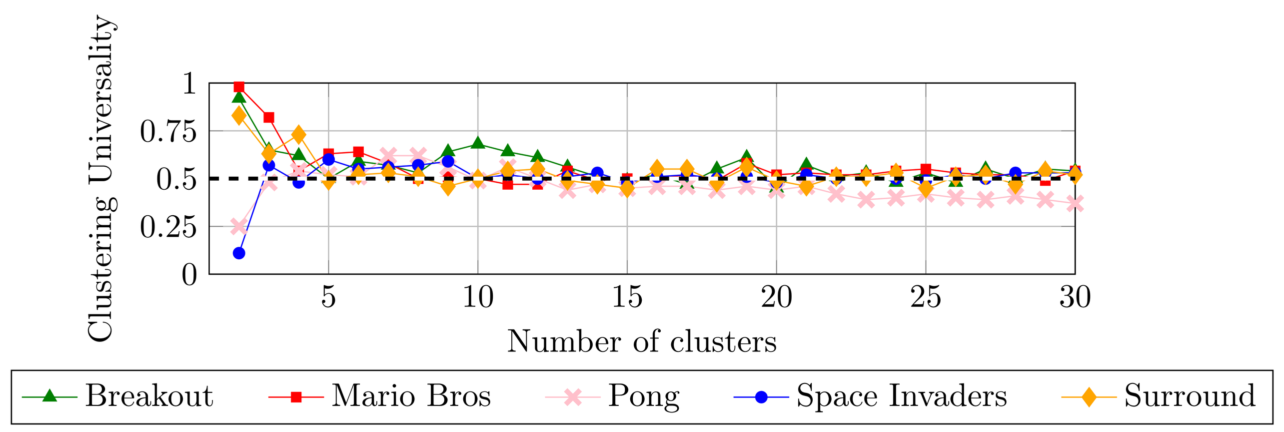


Institutional Partners

We gratefully acknowledge the support of our institutional partners. Their contribution and collaboration made this work possible.

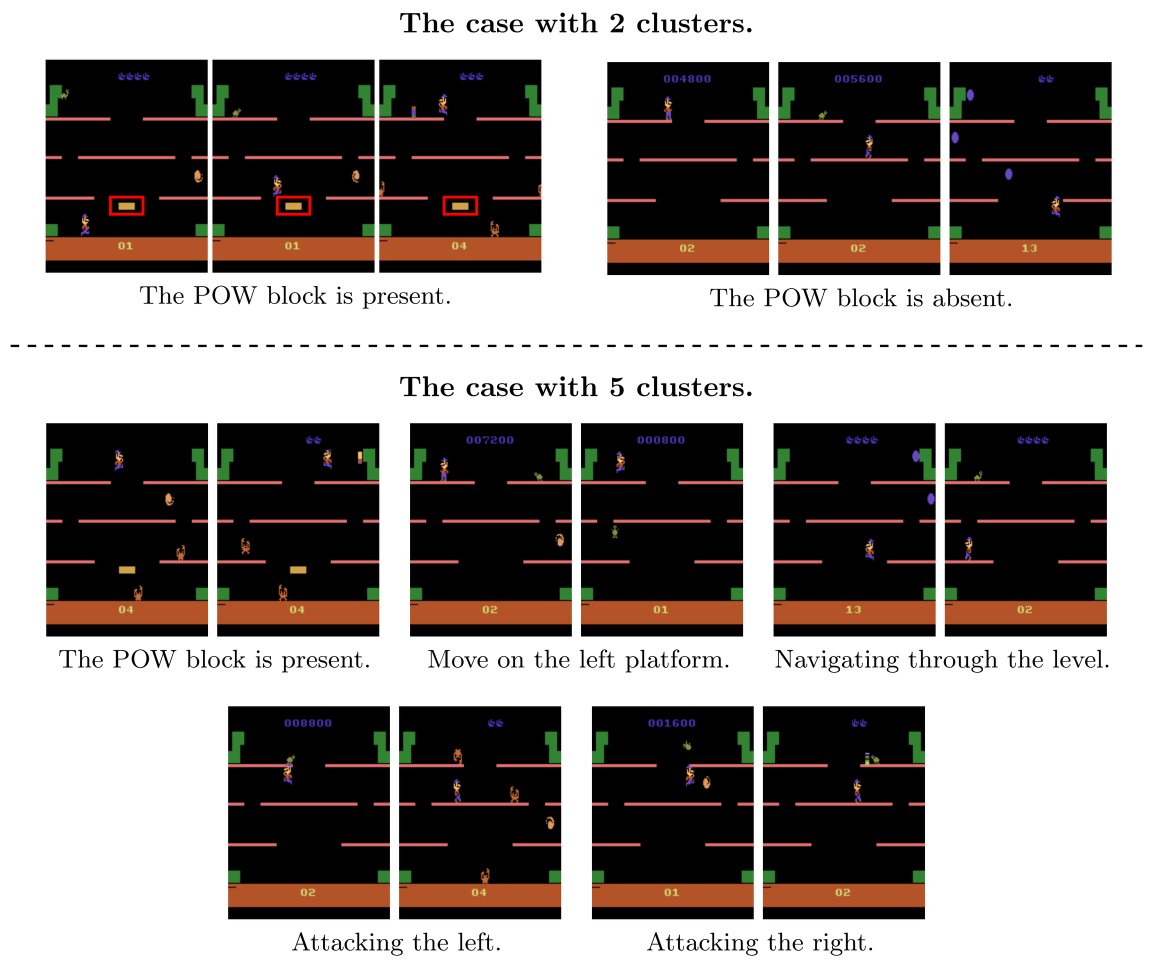
(5) Evolution Clustering Universality

We observe that regardless of the number of clusters chosen, the clustering remains universally good, with an Adjusted Rand Index of approximately 0.5. This indicates that the representations are stable across different learning processes.



(6) Semantic Granularity

When we adjust the number of clusters, we obtain representations at different levels of abstraction. A higher number of clusters tends to capture more tactical representations, while a lower number of clusters reveals more strategic representations.



Full Paper

For more details on the methodology and results, please refer to the full paper. Scan the QR code to access the complete publication.